

Score

1

Correct

Type : Mcq

Attempted

What is the output of the following code:

```
nums = [5, 10, 15, 20, 25]  
print(nums[-4:-1])
```

- ☐ [5, 10, 15]
- ☐ [10, 20, 25]
- ☐ [15, 20, 25]
- ☒ [10, 15, 20]



Correct Answer

Question 2 Of 30

Select Question

Q28

< Previous

Next: >

Score

2

Correct

Type : Mcq

Attempted

You need to display a report of all customers and their orders. Customers who have not placed any orders should also be shown with 'NULL' values in the orders column. Which query structure should you use?

- ☐ SELECT customers.customer_id, orders.order_id FROM customers
FULL JOIN orders ON customers.customer_id = orders.customer_id;
- ☐ SELECT customers.customer_id, orders.order_id FROM customers
RIGHT JOIN orders ON customers.customer_id = orders.customer_id;
- ☐ SELECT customers.customer_id, orders.order_id FROM customers
INNER JOIN orders ON customers.customer_id = orders.customer_id;
- ☒ SELECT customers.customer_id, orders.order_id FROM customers
LEFT JOIN orders ON customers.customer_id = orders.customer_id;



Correct Answer

Select Question

Question 3 Of 30

Q23

< Previous

Next: >

Score 2

Correct

Type : Mcq

Attempted

A retail chain is tracking real-time inventory updates, customer orders, and payments. The company requires frequent updates and quick query responses on individual transactions. Which system would best support these operations?

- ☐ Operational System
- ☐ NoSQL Database
- ☒ OLTP System
- ☐ OLAP System

✓ Correct Answer

Select Question

Question 4 Of 30

Q29

< Previous

Next: >

Score 2

Correct

Type : Mcq

Attempted

You have a dataset `df` with several columns, including one with sales data. You want to find the average of the sales column for only the rows where the product category is "Electronics." Which operation in Pandas would you use?

- ☐ `df.groupby('category')['sales'].mean()`
- ☐ `df['sales'].groupby('category').mean()`
- ☐ `df['sales'].mean()`

☒ `df[df['category'] == 'Electronics']['sales'].mean()`

✔ Correct Answer

Question 5 Of 30

Select Question

Q6

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

You want to simplify a long query on a sales table by first filtering high-value rows and then selecting from that filtered set. You decide to use a CTE named high_sales that returns rows where amount > 1000. After defining the CTE, you want to select all columns from it. Which SQL snippet uses a CTE correctly?

☒ WITH high_sales AS (SELECT * FROM sales WHERE amount > 1000)
SELECT * FROM high_sales;

✓ Correct Answer

☐ WITH high_sales (SELECT * FROM sales WHERE amount > 1000)
SELECT * FROM high_sales;

☐ SELECT * FROM sales WITH high_sales AS (amount > 1000);

☐ SELECT * FROM high_sales; WITH high_sales AS (SELECT * FROM sales WHERE amount > 1000);

Submitted

Status

35 / 40

Total Score

87.50%

Percentage

Passed

Result

16

Time Spent (mins)

Select Question

Question 6 Of 30

Q2

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

Which statement is true about nested if conditions in Python?

- ☐ Nested if statements are faster than normal if
- ☐ Nested if statements must end with an else
- ☐ Python does not allow nested if statements

☒ A nested if runs only if the outer condition is true



Correct Answer

Submitted

Status

35 / 40

Total Score

87.50%

Percentage

Passed

Result

16

Time Spent (mins)

Select Question

Question 7 Of 30

Q3

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

Which SQL keyword is used to count the number of rows in a result set?

☐ COUNT(ALL)

☒ COUNT(*)

☐ TOTAL

☐ SUM



Correct Answer

Submitted

Status

35 / 40

Total Score

87.50%

Percentage

Passed

Result

16

Time Spent (mins)

Select Question

Question 8 Of 30

Q5

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

Which query returns all employees with a salary greater than 50000?

- ☐ SELECT * FROM employees WHERE salary = 50000;
- ☐ SELECT * FROM employees HAVING salary > 50000;
- ☐ SELECT salary > 50000 FROM employees;

☒ SELECT * FROM employees WHERE salary > 50000;

✓ Correct Answer

What will be printed by the following code:

```
x = 10
if x > 5:
    print('Greater')
elif x == 10 print('Equal')
else:
    print('Smaller')
```

- ☐ Equal
- ☐ Smaller
- ☐ Error
- ☐ Greater

✓ Correct Answer

✗ Incorrect Answer

Question 10 Of 30 Select Question Q14 < Previous Next: >

Score 1 Correct Type : Mcq Attempted

How do you handle multiple types of exceptions in Python?

- ☐ Using multiple finally blocks
- ☐ Using multiple try blocks
- ☐ Using the error keyword
- ☒ Using multiple except blocks

✓ Correct Answer

Question 11 Of 30

Select Question

Q27

< Previous

Next: >

Score

2

Correct

Type : Mcq

Attempted

You want to find all products that are priced higher than the average price of products in the "Electronics" category. Which query correctly uses a subquery?

☒ `SELECT * FROM products WHERE price > (SELECT AVG(price) FROM products WHERE category = 'Electronics');`

✓ Correct Answer

☐ `SELECT * FROM products WHERE price > ALL (SELECT AVG(price) FROM products WHERE category = 'Electronics');`

☐ `SELECT * FROM products WHERE price >= (SELECT AVG(price) FROM products WHERE category = 'Electronics');`

☐ `SELECT * FROM products WHERE price = (SELECT AVG(price) FROM products WHERE category = 'Electronics');`

Score

2

Correct

Type : Mcq

Attempted

A manager needs to run a query to retrieve sales details but wants to avoid repeating the same subquery logic twice. The manager decides to use a CTE to simplify the query. Which pattern uses a CTE correctly?

☐ SELECT * FROM customer_totals JOIN orders o ON
customer_totals.customer_id = o.customer_id;

☐ WITH customer_totals AS (SELECT customer_id, SUM(amount) AS
total_amount FROM sales) SELECT * FROM customer_totals;

☒ WITH customer_totals AS (SELECT customer_id, SUM(amount) AS
total_amount FROM sales GROUP BY customer_id) SELECT
c.customer_id, c.total_amount, o.region FROM customer_totals c JOIN
orders o ON c.customer_id = o.customer_id;



Correct Answer

☐ WITH customer_totals AS (SELECT customer_id, SUM(amount) FROM
sales GROUP BY customer_id);

Question 13 Of 30

Select Question

Q18

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which of the following is the correct way to handle multiple exceptions in Python?

☐ Using one catch block

☒ Using different except blocks for different errors

✓ Correct Answer

☐ Using a single except block for all errors

☐ Using only the try block

Submitted

Status

35 / 40

Total Score

87.50%

Percentage

Passed

Result

16

Time Spent (mins)

Select Question

Question 14 Of 30

Q21

< Previous

Next: >

Score

2

Correct

Type : Mcq

Attempted

Which query adds a new column `email` to the `customers` table?

☒ ALTER TABLE customers ADD COLUMN email VARCHAR(100);

✓ Correct Answer

☐ UPDATE customers ADD email VARCHAR(100);

☐ ALTER customers ADD email VARCHAR(100);

☐ INSERT INTO customers ADD email VARCHAR(100);

Question 15 Of 30

Select Question

Q10

< Previous

Next: >

Score 0

Incorrect

Type : Mcq

Attempted

Which of the following statements is true when creating a table in SQL?

☐ You can create a table with only a column name but no data type.

☒ A table requires at least one column to be defined.

Correct Answer

☐ A table can be created without defining any columns.

Incorrect Answer

☐ A table's primary key can only be defined after the table is created.

Question 16 Of 30

Select Question

Q20

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

Which query returns all employees whose salary is greater than the average salary?

- ☐ SELECT * FROM employees WHERE salary = ALL (SELECT AVG(salary) FROM employees);
- ☒ SELECT * FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
- ☐ SELECT * FROM employees WHERE salary >= (SELECT AVG(salary) FROM employees);
- ☐ SELECT * FROM employees WHERE salary > ALL (SELECT AVG(salary) FROM employees);

✓ Correct Answer

Question 17 Of 30

Select Question

Q15

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which function is used to set the axis labels in a Matplotlib chart?

☐ plt.label_axes()

☒ plt.xlabel() and plt.ylabel()

✓ Correct Answer

☐ plt.set_labels()

☐ plt.axis_labels()

Submitted

Status

35 / 40

Total Score

87.50%

Percentage

Passed

Result

16

Time Spent (mins)

Select Question

Question 18 Of 30

Q4

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

Which code snippet correctly plots a simple line chart using Matplotlib?

- ☐ `plt.lineplot([1, 2, 3]); plt.show()`
- ☐ `plt.graph([1, 2, 3]); plt.show()`
- ☐ `plt.drawline([1, 2, 3]); plt.show()`
- ☒ `plt.plot([1, 2, 3]); plt.show()`



Correct Answer

Question 19 Of 30 Select Question

Q9

< Previous Next: >

Score 1 Correct Type : Mcq Attempted

What does unit testing focus on in software development?

- ☒ Testing individual components or functions
- ☐ Testing system performance
- ☐ Testing user interfaces
- ☐ Testing the entire application

✓ Correct Answer

Submitted

Status

35 / 40
Total Score

87.50%

Percentage

Passed
Result

16
Time Spent (mins)

Question 20 Of 30

Select Question

Q13

[← Previous](#)

Next: >

Score	1
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1

Correct

Type : Mcq

Attempted

To create a histogram in Matplotlib, which of the following functions should be used?

- ☐ plt.plot()
- ☐ plt.bar()
- ☐ plt.line()
- ☒ plt.hist()

✔ Correct Answer

What is the output of the following code:

```
x = 12

if x > 15:
    print("High")
elif x > 10:
    print("Medium")
else:
    print("Low")
```

☐ Error

☐ High

☒ Medium

☐ Low

✓ Correct Answer

Submitted

Status

35 / 40

Total Score

87.50%

Percentage

Passed

Result

16

Time Spent (mins)

Select Question

Question 22 Of 30

Q16

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

Which of the following code snippets correctly creates a line graph in Matplotlib?

☐ plt.display([1, 2, 3])

☒ plt.plot([1, 2, 3])



Correct Answer

☐ plt.plotgraph([1, 2, 3])

☐ plt.show_line([1, 2, 3])

Question 23 Of 30 Select Question Q8 < Previous Next: >

Score 1 Correct Type : Mcq Attempted

Which Python keyword is used after the try block to handle an exception?

☐ error

☒ except

✓ Correct Answer

☐ catch

☐ warn

Score

0

Incorrect

Type : Mcq

Attempted

You need to calculate the total sales for each product. The sales data includes a 'product_id' and 'amount' in the 'sales' table. You want to join it with the 'products' table to display the product name along with its total sales. Which query correctly achieves this?

- ☐ SELECT products.product_name, sales.amount FROM sales JOIN products ON sales.product_id = products.product_id;

- ☒ SELECT products.product_name, SUM(sales.amount) FROM sales JOIN products ON sales.product_id = products.product_id GROUP BY products.product_name;

- ☐ SELECT products.product_name, SUM(sales.amount) FROM sales LEFT JOIN products ON sales.product_id = products.product_id GROUP BY products.product_name;

- ☐ SELECT products.product_name, COUNT(sales.amount) FROM sales LEFT JOIN products ON sales.product_id = products.product_id GROUP BY products.product_name;



Correct Answer



Incorrect Answer

```
x = 7
y = 10

if x > y:
    print("Larger")
elif x == y:
    print("Equal")
else:
    print("Smaller")
```

- ☐ Error
- ☐ Equal
- ☐ Larger
- ☒ Smaller

✓ Correct Answer

Submitted

Status

35 / 40

Total Score

87.50%

Percentage

Passed

Result

16

Time Spent (mins)

Select Question

Question 26 Of 30

Q19

< Previous

Next: >

Score 0

Incorrect

Type : Mcq

Attempted

Which SQL query returns the average salary of employees in department 7?

☐ GROUP BY dept_id

☒ SELECT AVG(salary) FROM employees GROUP BY dept_id;

✗ Incorrect Answer

☐ SELECT salary FROM employees WHERE dept_id = 7;

☐ SELECT AVG(salary) FROM employees WHERE dept_id = 7;

✓ Correct Answer

Question 27 Of 30

Select Question

Q26

< Previous

Next: >

Score

2

Correct

Type : Mcq

Attempted

A report needs to display the top 3 highest-selling products and their total revenue. You decide to use a CTE to extract the total revenue per product and then filter to get the top 3. Which query structure would you use?

- ☐ SELECT * FROM sales GROUP BY product_id ORDER BY revenue DESC LIMIT 3;
- ☒ WITH top_products AS (SELECT product_id, SUM(revenue) AS total_revenue FROM sales GROUP BY product_id) SELECT * FROM top_products ORDER BY total_revenue DESC LIMIT 3;
- ☐ SELECT * FROM sales WHERE product_id IN (SELECT product_id FROM sales GROUP BY revenue);
- ☐ SELECT product_id, SUM(revenue) AS total_revenue FROM sales GROUP BY product_id ORDER BY revenue DESC LIMIT 3;

✓ Correct Answer

Question 28 Of 30

Select Question

Q22

< Previous

Next: >

Score 2

Correct

Type : Mcq

Attempted

A large e-commerce platform collects user activity logs, including clicks, search queries, and purchase history. The company wants to analyze this data to generate insights about user behavior across different time periods. Which part of the data architecture is most suitable for this type of analysis?

- ☐ OLTP Database
- ☐ NoSQL Document Store
- ☐ Data Lake
- ☒ Data Warehouse

✔ Correct Answer

Score

1

Correct

What is the output of this Python code:

```
try:
    result = 5 / 'a'
except TypeError:
    print('Cannot perform operation with string')
```

- ☐ Error
- ☐ Nothing
- ☐ TypeError occurs and is handled
- ☒ Cannot perform operation with string



Correct Answer

Question 30 Of 30

Select Question

Q1

< Previous

Score 1

Correct

Type : Mcq

Attempted

What is the result of `len([10, 20, 30, 40])`?

☐ 3

☒ 4

☐ Error

☐ [10, 20, 30, 40]

✓ Correct Answer